

Installation Instructions

This set of instructions will create a 3 partition version of the architecture all on one disk. See the end of the document for details on how to create the various partitions on different disks Be **very** careful to chose the right removable disk when running the install program.

You will need

- A functioning **Linux** operating system which has access to USB devices. For Windows users you cannot use a Hyper-V vm, nor WSL as these do not have USB access. The best solution is to install Ubuntu on a usb key and boot it into "Try Before You Install" mode.
- A USB key or a microSD card in a USB holder of at least 8GB capacity. If this is to be your main multiboot repository it is recommended be at least 128GB, depending upon the number of OS that you wish to install.

Installation Steps

Base USB Key

1. Download the Pi_MultiBoot.tar.gz file from my repository and unpack it.

```
cd ~
wget https://github.com/AndyDeSheffield/Pi_Multiboot/releases/download/latest/Pi_Multiboot.tar.gz
tar -xzf Pi_Multiboot.tar.gz
```

1. plug in the target USB key if you haven't already done so
2. Identify your target USB key **carefully** using the multiboot-build.sh -l command. Note that minimal help is available with multiboot-build.sh -h

```
cd ~/Pi_Multiboot/create_structure
./multiboot-build.sh -h
./multiboot-build.sh -l
```

1. Identify the device that you wish to image here I'll assume it is **/dev/sdb** (replace as necessary) and that we are building for a pi4 (again replace as necessary run with -h to see models) Then for peace of mind I suggest that you do a dry run of the creation process using the -n option Also we will stick to the default partition sizes

```
./multiboot-build.sh -n -t=/dev/sdb -m=pi4 -b -s -i
```

1. If you are happy then rerun for real. This needs sudo privileges

```
sudo ./multiboot-build.sh -t=/dev/sdb -m=pi4 -b -s -i
```

1. You should now have a 3 partition USB key.

Admin OS Image

For convenience I've provided a prepared admin os based on Raspi_Lite but with a minimal Xserver and console. There is a pre-installed grub entry that boots it. If you want to install this (recommended) continue on.

Automated Installation

1. Run the included installation script, replacing /dev/sdb with your newly installed disk identifier (the -l options lists current disks for a safety check)

```
cd ~/Pi_Multiboot/create_structure
./install_multiboot_admin_os.sh -l
```

Then when happy (choosing the right disk)

```
./install_multiboot_admin_os.sh /dev/sdb
```

Manual Installation

If for any reason the automated installation didn't work or if you prefer to know what's going on here are the manual Steps (Assumes IMAGES partition is /dev/sdb3)

1. Mount the IMAGES partition of the key

```
sudo mkdir -p /mnt/realimages
sudo mount /dev/sdb3 /mnt/realimages
cd /mnt/realimages/staging/
```

1. Fetch the admin tarball, unpack it and install the image directory (update the release directory and name as necessary)

```
wget https://github.com/AndyDeSheffield/Pi_Multiboot/releases/download/latest/MultiBoot_Admin_os.tar.gz
tar -xzf MultiBoot_Admin_os.tar.gz -C /mnt/realimages
sync
```

1. Unmount the partition and optionally remove the mountpoint

```
cd ~
sudo umount /mnt/realimages
sudo rmdir /mnt/realimages
```

Testing

Transfer the usb key to the target device and try to boot. If it is a microSD card you can optionally put it into the slot on the pi. **Note that the BOOT partition on this key must be the only one in the system.** Also only one SYSTEM partition is allowed. You can have any number of IMAGES partitions

For the MultiBoot Admin OS, User=pi, Password=pi

Split Disks

Notes on splitting the architecture over different disks. This quite simply comes down to repeating steps 2 to 6 of the BASE USB section of this guide but specifying less partitions at stages 4 and 5. For example to just put the Boot partition on a disk run

```
sudo ./multiboot-build.sh -n -t=/dev/sdb -m=pi4 -b
sudo ./multiboot-build.sh -t=/dev/sdb -m=pi4 -b
```

Then for a second disk with the SYSTEM and IMAGES partitions

```
sudo ./multiboot-build.sh -n -t=/dev/sdb -m=pi4 -s -i
sudo ./multiboot-build.sh -t=/dev/sdb -m=pi4 -s -i
```